
Trustworthy AI and Causal Inference for Token Launch Platforms: Evidence, Stakeholders, and Market Design

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Token launch platforms sit at the intersection of digital markets, decentralized
2 infrastructure, and AI-assisted decision making, creating new challenges for evalu-
3 ating causal and stakeholder claims. We develop an interdisciplinary framework
4 that combines platform economics, causal econometrics, provenance-aware data
5 infrastructure, and trustworthy AI evaluation through a common evidence con-
6 tract. Using a cross-chain dataset covering Pump.fun and PumpSwap on Solana,
7 Clanker on Base, Four.meme on BNB Chain, and SunPump on TRON, we preserve
8 chain-specific evidence boundaries rather than treating heterogeneous platforms
9 as directly comparable. Three empirical applications and five platform-calibrated
10 known-truth experiments show how economically meaningful operational, predic-
11 tive, and mechanical conclusions can remain valid even when aggregate causal
12 or welfare effects are not identified. We then test whether language models re-
13 spect these boundaries through a 714-call PumpSwap audit and a 60-call Telegram
14 evaluation across three backends. Few-cluster disclosure is associated with sub-
15 stantially lower unsafe causal affirmation and higher correct-boundary responses,
16 while the Telegram case shows consistent preservation of a predictive-not-causal
17 boundary. The contribution is a cross-disciplinary approach to trustworthy analysis
18 of emerging digital markets that connects economic theory, causal identification,
19 reproducible data design, and AI evaluation within a single evidentiary framework.

20 1 Introduction

21 When language models interpret observational platform data, they can reproduce a reported pattern
22 while overstating what it establishes for affected stakeholders. Evaluating such output therefore
23 requires criteria beyond answer accuracy [Wang et al., 2023, Huang et al., 2024]. Token launch plat-
24 forms provide a demanding case because they automate issuance and connect initial price discovery
25 to secondary liquidity at scale. The released Solana cohort contains 832,941 terminal outcomes from
26 May 8 to June 10, 2026 and only 1,651 graduations; Pump.fun surpassed \$1 billion in cumulative
27 protocol revenue by early 2026 [DefiLlama, 2026]. These aggregates establish scale, not who benefits
28 or whether a platform rule caused the observed change. Because AI-assisted interpretations can
29 influence retail decisions, creator expectations, governance, and regulation, unsupported causal or
30 welfare claims may misstate who benefits and who bears risk.

31 Platform theory explains token adoption, financing, and two-sided market design [Cong et al., 2021,
32 Sockin and Xiong, 2023, Gryglewicz et al., 2021, Malinova and Park, 2023, Rochet and Tirole,
33 2003]. Empirical claims nevertheless often rely on aggregate before-and-after comparisons, such as
34 early PumpSwap volume [Cointelegraph, 2025]. Such comparisons neither establish a counterfactual
35 nor identify the stakeholder outcome to which a change should be attributed. Under staggered

36 adoption, pooled two-way fixed effects can also use inadmissible comparisons and negative weights
 37 [Goodman-Bacon, 2021, de Chaisemartin and D’Haultfœuille, 2020]. Language-model output can
 38 inherit this ambiguity. We call this unsupported upgrade *evidentiary escalation*. Auditing research
 39 motivates traceable evidence and explicit failure criteria [Raji et al., 2020, Suresh and Gutttag, 2021],
 40 but an applied audit still needs a reference boundary that separates supported operational statements
 41 from unsupported causal or welfare claims.



Figure 1: The four-chain release preserves source boundaries. The evidence contract combines the L0–L7 ladder with known-truth tests, and the audit evaluates boundary adherence.

43 Therefore, we treat each causal statement as a claim whose support depends on its data, design,
 44 uncertainty, and stakeholder. The evidence contract maps these requirements to the narrowest
 45 warranted interpretation and supplies the reference boundary for the model audit. Specifically, we
 46 concentrate on two linked research questions.

- 47 • **Q1: Which stakeholder claims about token launch platforms remain supportable under a**
 48 **common evidence contract?** Three applications trace the four-chain evidence base to final claim
 49 boundaries, while five known-truth experiments clarify the statistical gates. The resulting support
 50 is operational, predictive, or mechanical rather than aggregate causal or welfare.
- 51 • **Q2: Do language-model outputs preserve these claim boundaries as evidence is disclosed?**
 52 A full-factorial blind audit applies the frozen PumpSwap evidence object to three backends over
 53 714 calls, and a separate 60-call targeted Telegram evaluation tests a second claim boundary.
 54 PumpSwap disclosure effects are model dependent; the Telegram categorical scores are at ceiling
 55 under both registered conditions.

56 Figure 1 summarizes the progression from coverage-aware data to bounded interpretation.¹ After
 57 defining source coverage and the L0–L7 contract, the paper applies them to the real cases. The
 58 known-truth experiments examine the validity gates before the model audit tests adherence to the
 59 resulting boundary. Table 4 records each focal claim and its final status. The contract is an organizing
 60 device rather than a new estimator or a universal welfare assessment.

61 2 Related Work

62 Platform economics and market design define the relevant stakeholders and mechanisms. Price
 63 structure coordinates interdependent sides [Rochet and Tirole, 2003]. Market rules govern thickness,
 64 congestion, and safety [Roth, 2008], while token models extend these ideas to adoption, membership,
 65 and financing [Cong et al., 2021, Sockin and Xiong, 2023, Gryglewicz et al., 2021]. This makes
 66 stakeholder disaggregation part of identification rather than a reporting choice: the same platform
 67 rule can increase aggregate activity while leaving creator, trader, or community outcomes unresolved.

¹ Anonymous code and reproduction artifact: <https://anonymous.4open.science/r/Web3AI4I0-6605/>.

Table 1: Representative blockchain datasets and evaluation scope. Columns denote launchpad lifecycle (LP), multiple chains (MC), onchain and offchain linkage (OC), provenance or coverage states (PC), explicit event-validity representation or evaluation (EV), known causal truth (KT), and language-model audit (AA). A cross denotes a capability outside the evaluated scope, not a defect; a partial mark denotes bounded support.

Dataset or framework	Data scope			Claim validity			AI
	LP	MC	OC	PC	EV	KT	AA
Chartalist [Shamsi et al., 2022]	✗	✓	✗	✗	✗	✗	✗
EX Graph [Wang et al., 2024]	✗	✗	✓	✗	✗	✗	✗
Uniswap Indices [Chemaya et al., 2025]	✗	✓	✗	●	✗	✗	✗
Multi Chain GoG [Luo et al., 2024]	✗	✓	✗	✗	✗	✗	✗
XChainDataGen [Augusto et al., 2025]	✗	✓	✗	●	✗	✗	✗
FinML Chain [Chen et al., 2026]	✗	✗	✓	✗	●	✗	✗
MemeTrans [Hu et al., 2026]	✓	✗	●	✗	✗	✗	✗
Our Study	✓	✓	●	✓	✓	✓	✓

LP: launchpad lifecycle; MC: multiple chains; OC: onchain and offchain linkage; PC: explicit provenance or coverage states; EV: explicit representation or evaluation of event validity; KT: known causal truth; AA: language-model evidence audit. ✓ full support; ● bounded support; ✗ outside the evaluated scope. “Known truth” means calibrated evaluation with a known causal target, rather than predictive labels or observational outcomes. EV denotes the capability to record or assess event validity, not that every event is causally valid.

Research on blockchain evidence infrastructure shows that reusable cross-chain representations are feasible. Existing datasets cover transaction graphs, hierarchical token graphs, decentralized-exchange activity, social links, and offchain sentiment [Shamsi et al., 2022, Luo et al., 2024, Chemaya et al., 2025, Wang et al., 2024, Chen et al., 2026]. MELT is closest in task scope, studying Pump.fun transactions and post-migration risk within a migrated-token subset [Hu et al., 2026]. Within the comparison summarized in Table 1, these resources do not bind source coverage to the validity boundary of a launchpad intervention claim. A harmonized representation can therefore remain causally non-comparable when platforms differ in denominators, observation horizons, or available outcome layers.

Trustworthy language-model evaluations assess robustness, safety, and utility beyond task accuracy [Wang et al., 2023, Huang et al., 2024]. Algorithmic-auditing research emphasizes traceable evidence and lifecycle sources of harm [Raji et al., 2020, Suresh and Guttag, 2021]. These perspectives motivate explicit criteria but do not define causal claim boundaries for partially observed interventions.

Causal methods address pooled staggered two-way fixed effects, pretrends, and few clusters [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024, Rambachan and Roth, 2023, MacKinnon and Webb, 2018]. CLADDER evaluates language models against oracle answers for associational, interventional, and counterfactual queries [Jin et al., 2023]. Our audit complements that setting by testing adherence to a prespecified applied boundary under controlled disclosure. Neither an estimator nor a model evaluation repairs invalid timing, an uncollected outcome, or a welfare claim formed from aggregate activity.

3 Data

3.1 Construction

The public data release is organized around three layers: immutable inputs, canonical tables, and deterministic builders. Nine common tables provide a shared structure where applicable, while an event registry and an event-evidence table record intervention-specific information. Manifests document provenance, scope, and output integrity, and the builders enforce consistency across keys, references, counts, and coverage states. The resulting schema supports cross-chain extension without assuming that different platforms provide equivalent evidence. In particular, coverage is retained at the level of the claim so that differences in what was observed do not become hidden differences in the effective comparison sample.

Missingness is defined relative to how a record could have been observed and what the focal claim requires. The release therefore distinguishes genuine absence from unacquired, inapplicable, or intentionally excluded information, preventing a finite-snapshot nonobservation from being treated as

Coverage remains chain specific after standardization

	Launch universe	Pool and liquidity	Terminal outcome	Activity outcomes	Metadata	Event design status
Solana Pump.fun	832,941 outcomes	1,651 graduated	terminal state	294 decoded tokens	outside release	conditional
Base Clanker	62,618 launches	62,618 pools	not collected	not collected	outside release	accepted
BNB Chain Four.meme	1,593,679 launches	15,403 pools	not collected	not collected	outside release	rejected
TRON SunPump	104,548 launches	1,831 pools	not collected	not collected	outside release	rejected

■ Observed ■ Bounded or selected
■ Not collected ■ Excluded by release policy

Figure 2: A common schema preserves rather than erases chain-specific evidence boundaries. Cells distinguish observed, bounded or selected, uncollected, and release-excluded layers. Event-design status is a separate identification assessment: accepted, conditional, and rejected indicate whether a registered intervention passes the activation-evidence and comparison-design gates. These labels do not grade data quality or remove a chain from the release. All four chain cores contribute to the standardized cross-chain evidence map and support descriptive or mechanism claims within their declared coverage. Counts describe the released snapshot and are not pooled causal denominators.

Table 2: Canonical v1 scope. All boundaries are reported as UTC observation windows. Denominators represent different protocol entities and are not direct cross-chain comparison samples.

Chain and platform	Core units	Lifecycle units	Observation window (UTC)
Solana , Pump.fun	832,941 outcomes	1,651 graduations	2026-05-08 to 2026-06-10
Base , Clanker	62,618 launches	62,618 pools	2025-08-18 to 2025-10-01
BNB , Four.meme	1,593,679 launches	15,403 pools	2024-07-03 to 2026-08-12
TRON , SunPump	104,548 launches	1,831 pools	2024-08-09 to 2026-08-11

a structural zero. Figure 2 shows how these distinctions produce chain-specific evidence boundaries after standardization. Event-design status is reported separately because it concerns whether an intervention supports a credible causal comparison, not how complete the underlying chain data are. Appendix A provides the schema fields, acquisition procedures, and validation rules.

3.2 Chain Scope

Table 2 summarizes the four chain cores. Solana supplies full terminal outcomes and selected post-graduation trading layers. Base, BNB Chain, and TRON instead supply launch universes with bounded lifecycle coverage. Official application programming interfaces enrich metadata but never define a universe when pagination is capped. The unified schema and explicit observation windows establish the scope of cross-chain analysis. The chains can share construction and field meaning, but their different denominators and outcome layers do not form a pooled causal panel. These scope differences also determine which claims the evidence can support. Detailed chain-specific scope appears in Appendix A.

4 Methodology

4.1 Evidence Contract

A platform claim is evaluated as an evidence contract rather than a point estimate. A complete claim identifies whose outcome is measured and states the intervention, comparison rule, horizon, estimator, uncertainty procedure, and allowed interpretation. This structure first separates stakeholder outcomes. Platform activity is not a welfare scalar because a price or rule change can subsidize one side while monetizing another [Rochet and Tirole, 2003], and market design distinguishes thickness from congestion and safety [Roth, 2008]. Creator outcomes therefore describe participation,

graduation, time to graduation, and gross fee receipts. Trader outcomes capture breadth, persistence, concentration, execution quality, and measurable harm. Platform outcomes concern fee revenue and sustained activity, while community and host-ecosystem outcomes cover inclusion, security, and spillovers. A result can be positive for one stakeholder and unidentified or negative for another. The resulting assessment ends with a stakeholder claim vector rather than an overall verdict. Appendix C defines the complete metric set and interpretation boundaries.

The evidentiary meaning of these outcomes depends on source depth and temporal resolution. Market, token, transaction, wallet, and social layers answer different economic questions. Aggregate indices measure platform activity and decentralization at a common frequency [Chemaya et al., 2025]. Linked social and transaction data can instead reveal attention or entity relationships that remain invisible onchain [Wang et al., 2024, Chen et al., 2026]. Aggregate volume may be complete but economically coarse, whereas decoded token outcomes may be rich but selected. Calendar aggregation can also change exposure to the intervention, which makes frequency part of the estimand. Every result is therefore paired with its source boundary, unit, horizon, frequency, and coverage state.

Identification completes the contract by fixing the treatment family, target population, outcome, comparison rule, and estimand before the effect sign is inspected. The protocol requires activation evidence, anticipation and pretrend diagnostics, admissible controls, and design-appropriate uncertainty. Under staggered adoption, group-time or imputation estimators replace pooled two-way fixed effects when their assumptions are credible [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Borusyak et al., 2024]. Repeated interventions require clean event windows or an intertemporal estimand [de Chaisemartin and D’Haultfœuille, 2026]. With few platform clusters, inference must disclose its attainable resolution [MacKinnon and Webb, 2018]. Sensitivity analysis records departures from parallel trends rather than treating a failed diagnostic as validation [Roth and Sant’Anna, 2023, Rambachan and Roth, 2023]. No estimator repairs an invalid event, contaminated comparison, or missing stakeholder outcome, so the contract retains “not identified” as a valid result.

For the common-date PumpSwap intervention, the dynamic diagnostic takes the form

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k D_i \mathbb{I}\{t - T = k\} + \varepsilon_{it}, \quad (1)$$

where D_i identifies the treated protocol and week -1 is omitted. The pre-event coefficients assess the stability of the comparison design. The post-event coefficients admit a causal interpretation only when the timing and comparison assumptions hold.

4.2 Evidence Ladder

The L0 to L7 ladder operationalizes the evidence contract. L0 captures the dashboard style treated before and after estimand. Each subsequent rung adds a comparison or validity gate. The ladder is not a claim that each level uniquely identifies the marginal contribution of one component. Components can interact, and their order is fixed for auditability rather than factorial attribution. Table 3 defines the common contract used by the applications.

4.3 LLM Evidence-Boundary Evaluation

For evidence state E , let $B(E)$ denote the strongest claim permitted by the frozen reference analysis and $C_m(E)$ the claim produced by backend m . Escalation occurs when $C_m(E)$ exceeds $B(E)$, as when a predictive association is presented as causal. Boundary adherence requires more than abstention. A response must reject escalation while retaining warranted operational or predictive claims.

Controlled disclosure fixes the empirical question, prompt scaffold, and scoring rule. Table 5 in Appendix F defines the evidence content and factor status of M0–M7. With M0–M3 fixed, M4–M7 form a complete 2^4 design with controls for answer leakage and absent evidence. The design tests evidence-conditioned interpretation, not causal reconstruction from raw blockchain data.

PumpSwap scores unsafe causal affirmation, correct boundary, and operational-claim retention. Evidence-ID and missing-slot measures record module use. Telegram retains the three backends, zero temperature, and scaffold family while separating causal from predictive status. Model-reported confidence is exploratory. The evaluation comprises one full-factorial case and one targeted second case within one scaffold family.

Table 3: The L0 to L7 claim validity ladder. Each rung adds evidence or a validity gate while preserving the research claim whenever the data permit. Deterministic code evaluates statistical behavior; the model-evaluation arm tests whether a language model respects the same evidence boundary.

Level	Evidence added	Question answered	Failure made visible
L0	Treated before-and-after summary	Did the outcome change?	No comparison, estimand, or stakeholder boundary
L1	Explicit comparison rule	Did treated and comparison outcomes diverge?	Interference, concurrent shocks, inadmissible controls
L2	Unit and time adjustment	Is the contrast robust to unit and time factors?	Functional-form and aggregation dependence
L3	Cohort and event-time structure	Are timing and heterogeneous exposure represented?	Forbidden comparisons and estimand mismatch
L4	Timing, anticipation, and pretrend gates	Does nominal timing match economic activation?	Misdated treatment and unstable counterfactual trends
L5	Fixed horizons and sequential-event rules	Are exposure and later interventions aligned?	Censoring and overlapping interventions
L6	Design-appropriate uncertainty	Does inference respect dependence and cluster count?	Undercoverage, coarse exact tests, low power
L7	Stakeholder claim vector and sensitivity	Which claims remain supported, for whom?	Aggregate activity presented as welfare or inclusion

Table 4: Claim–evidence–stakeholder map. Status denotes the narrowest supported claim, not an aggregate platform verdict. Numerical cells are generated from archived artifacts.

Claim evaluated	Affected stakeholder	Required evidence and assumptions	Available evidence and status
Market migration effect	Operators; creators; traders	Protocol-day outcome, activation, DEX controls, stable window, valid inference	Daily/TWFE estimate 0.412, CI $[-0.128, 0.951]$; pretrend risk and four-cluster exact $p = 0.6875$. Not identified.
Post-migration venue operation	Creators and traders	Token-level activity, fixed horizon, explicit coverage-complete windows	RPC: 1,636/1,651 lower-bound active tokens; 762 complete windows, all active. Mechanism claim supported.
Social signal and retail risk	Token creators; communities; retail traders	Matched terminal outcomes, timing falsifications, holder snapshots, and an exogenous shock for causality	Telegram association: 0.945 percentage points, CI $[0.732, 1.163]$, E-value 5.02; no qualifying shock. Concentration proxy: 52.4 percentage points higher risk. Predictive/proxy, not causal.
Creator-fee rule incidence	Creators; traders; operators	Activation transaction, vault transfer, admissible control, stakeholder welfare	Verified +10, 732 lampport vault transfer; Pump–Moonshot launch diagnostic -0.966 , CI $[-1.252, -0.680]$; no accepted control. Mechanical incidence supported; welfare not identified.
AI evidence following	Reviewers and users of AI-assisted analysis	Blind prompts, factorized disclosures, model/version, repeated runs, controls, prespecified scoring	3 returned models; 714/714 registered calls succeeded; blind cumulative and 2^4 factorial evidence design; parsed structured outputs and per-call scores archived. Bounded, heterogeneous evaluation.

172 4.4 Known-Truth Stress Tests

173 Real applications reveal whether conclusions change but do not provide causal ground truth. We
174 therefore add five semi-synthetic stress tests calibrated from active platform data. Real data determine
175 untreated structure or selection-relevant slopes; synthetic assignment and effects provide known truth.
176 Methods are scored against the same estimand and generated panel. The tests isolate staggered hetero-
177 geneity, activation timing, four-cluster inference, latent-growth adoption, and temporal aggregation.
178 They establish the statistical role of the validity gates, while the model audit remains tied to the frozen
179 PumpSwap case. Their conclusions apply only to the calibrated data-generating processes.

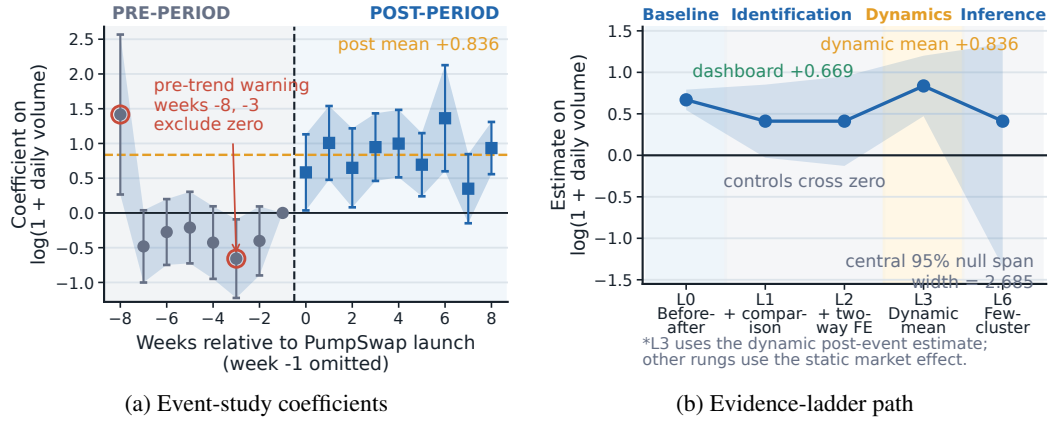


Figure 3: PumpSwap evidence diagnostics. Panel (a) shows positive postperiod volume estimates alongside unstable preperiod coefficients. Week minus one is omitted, and shaded regions report the displayed uncertainty intervals. Panel (b) shows how the initially positive dashboard conclusion becomes uncertain once comparison design and few-cluster inference are introduced. At L6, shading reports the central 95% null-reference span from all 16 Rademacher sign assignments, not a population confidence interval. Together the panels support a diagnostic description rather than an identified causal migration effect. The final stakeholder-specific claim at L7 is reported in the text and Table 4.

5 Empirical Analysis

The empirical analysis applies the contract to three applications and known-truth calibration. The cases address market-level causal attribution, token-level prediction, and mechanical fee incidence. Full designs and simulation protocols appear in Appendices D and E.

5.1 Claim Robustness

The applications trace how timing, comparison design, uncertainty, and stakeholder scope change a platform conclusion. Validity checks can weaken a causal interpretation while preserving a narrower descriptive, predictive, or mechanical result.

PumpSwap provides the market-level application using protocol-day observations over a ± 90 -day window around March 20, 2025. Pump ecosystem volume is compared with screened Raydium, Orca, and Meteora series. The intervention is the registered launch date, supported by public evidence but not a verified activation transaction. The outcome is $\log(1 + \text{daily volume})$, and the estimand is the change relative to the screened controls. The design progresses from a before-and-after contrast to fixed effects, a dynamic event study, and exact inference over four clusters. Causal interpretation requires valid timing, stable comparisons, and regular pre-event behavior.

The before-and-after contrast is 0.669 log points. With controls and two-way fixed effects it falls to 0.412, with a 95 percent confidence interval of $[-0.128, 0.951]$. Pre-event instability in Figure 3a and an exact Rademacher p value of 0.6875 preclude an affirmative causal conclusion. Token-level evidence supports a narrower mechanism claim. Of 1,651 graduated tokens, 1,636 show observed 30-day transaction-proxy activity, and all 762 complete windows are active. PumpSwap is an operational post-migration venue for the observed tokens, but the causal migration effect, execution quality, and trader welfare remain unidentified.

The token-level application examines Telegram metadata in 832,941 terminal Solana outcomes. Exposure is measured at launch, the outcome is graduation, and comparison units are formed within launch-day and covariate-matched cells. Launch-day fixed effects, coarsened exact matching, and clustered bootstrap uncertainty estimate the association. Outcome-timing falsification and six candidate public shocks probe endogenous signal adoption before any audience response.

Graduation is 1.485 percent among tokens with Telegram metadata and 0.166 percent without it. Matching yields a 0.945 percentage-point difference with a launch-day clustered interval of $[0.732, 1.163]$ and an E-value of 5.02. Yet the association is already present in the five-minute outcome and disappears after 60 minutes, while no screened shock supplies exogenous exposure.

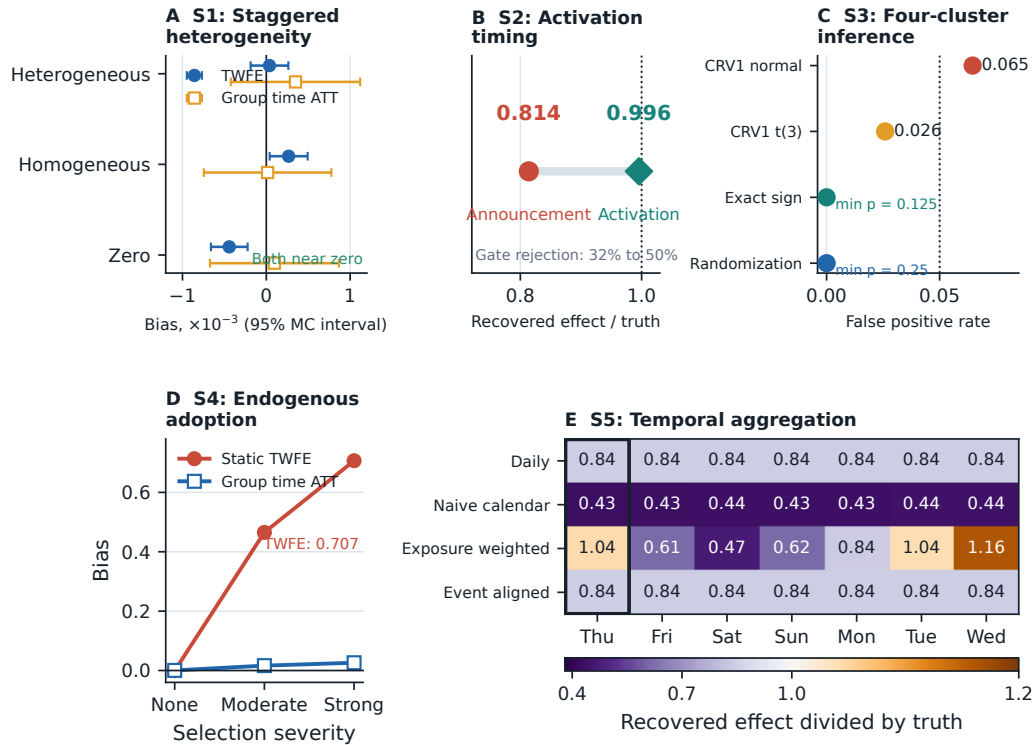


Figure 4: Five calibrated stress tests locate distinct claim-validity boundaries. Panels A and D examine heterogeneous effects and adoption selection. Panels B, C, and E examine timing miscoding, four-cluster calibration, and temporal aggregation. Panel A reports 95 percent Monte Carlo intervals.

Telegram metadata is therefore predictive and consistent with preexisting quality or attention, not an identified treatment.

The rule-incidence application separates program support from economic activation. A May 12 transaction records no creator-vault transfer, while onchain evidence verifies a 10,732-lamport vault increase on May 13, 2025. Concurrent changes and displacement leave no admissible platform control. The result supports gross mechanical incidence, not creator net welfare or a platform-wide effect. The remaining diagnostics are descriptive, and the bounded Base reconstruction validates the event pipeline only. Appendix B reports the complete estimates.

5.2 Known-Truth Calibration

Five Monte Carlo experiments combine platform-calibrated untreated structure with synthetic assignment and effects. Each isolates a different source of claim failure, so no single result implies general robustness. Figure 4 places the results on a common evidence map.

Specification diagnostics depend on the realized design. Across eight Clanker v4.1 cohorts, two-way fixed effects and a group-time estimator are approximately unbiased when cohort modifiers and effective weights align. Negative weights indicate risk rather than automatic bias. Timing error is distinct. In the observed five-day gap, coding the announcement recovers 81 percent of the known effect, whereas verified activation recovers essentially all of it. The short-sample pre-event gate still rejects in 32 to 50 percent of zero-effect or no-anticipation replications, so verified timing does not make the gate reliable.

Inference and identification impose different limits. With four clusters, exact sign and assignment procedures avoid asymptotic overstatement, but minimum two-sided p values of 0.125 and 0.25 preclude five-percent rejection. Under latent adoption selection, the group-time estimator reduces bias relative to two-way fixed effects, although nominal coverage remains incorrect because the identifying assumption fails.

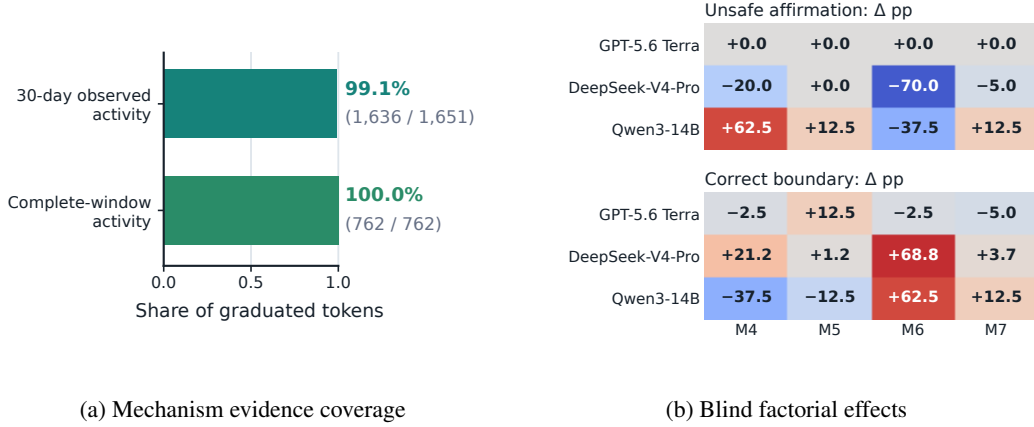


Figure 5: PumpSwap claim-boundary diagnostics. Panel (a) reports mechanism coverage, and panel (b) reports percentage-point matched-cell effects for M4–M7. Appendix F gives the intervals and call-level sensitivity analysis.

Naive weekly estimates for Thursday PumpSwap recover 43–44% of a transient target and 47–48% of a persistent target. Exposure weighting helps with persistence but not a short response after temporal information is lost. Temporal aggregation therefore changes the estimand.

5.3 LLM Boundary-Adherence Audit

The PumpSwap arm contains 160 calls per backend across 16 M4–M7 conditions, with 10 zero-temperature repeats per cell and fixed M0–M3 evidence, scaffold, and scoring. Unsafe causal affirmation is 0.0% for GPT-5.6 Terra, 35.0% for DeepSeek-V4-Pro, and 56.3% for Qwen3-14B. Their correct-boundary rates are 88.8%, 64.4%, and 31.3%, respectively.

Across 2,000 fixed-design hierarchical draws, M6 is associated with a 35.8-point decrease in unsafe affirmation, with a descriptive 95% interval of $[-48.8, -23.3]$, and a 42.9-point increase in correct-boundary rates, with an interval of $[29.2, 55.8]$. It is the only module with a nonpositive unsafe contrast for every backend, while M4, M5, and M7 do not move both scores consistently. Under complete M0–M7 disclosure, GPT and DeepSeek return the correct boundary in 10/10 calls. Qwen does so in 0/10 and affirms causality in 10/10. The intervals describe the fixed panel, not a model population. Appendix F reports full contrasts and call-level sensitivity analysis.

Telegram tests a predictive-to-causal boundary with the same backends and scaffold. T0 supplies the observed and matched associations, clustered interval, E-value, and matched support. T1 adds five-minute falsification, the 60-minute result, the 0/6 qualifying-shock screen, and unavailable or proxied controls. All cells record 0/10 unsafe affirmations and 10/10 for both boundary and predictive-association retention. The unchanged scores indicate a ceiling rather than a disclosure effect or broad generalization.

6 Conclusion

This paper develops a provenance-aware evidence contract connecting source coverage, causal design, stakeholder interpretation, and model evaluation. Across four chains, three applications and five known-truth experiments show that useful operational, predictive, or mechanical conclusions can remain after unsupported causal or welfare claims are rejected. The contract therefore treats “not identified” as an informative result without discarding supported evidence.

In the model audit, PumpSwap varies by backend, with M6 the only module producing a nonpositive unsafe-affirmation contrast for all three. Telegram is at ceiling while preserving the predictive rather than causal boundary. These results favor testing responses to specific validity conditions rather than context volume. Future work can examine whether provenance-linked retrieval or interactive analysis improves adherence without suppressing supported claims.

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Appendix

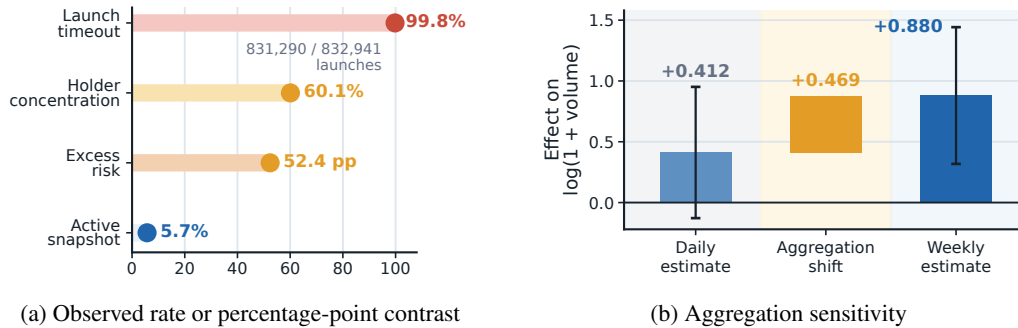


Figure 6: Application diagnostics for stakeholder denominators (a) and temporal aggregation (b).

A Data and Coverage

The release separates immutable source artifacts, canonical outputs, and deterministic builders. Its common schema covers tokens, launches, protocol configurations, pools, liquidity initializations, life-cycle events, metadata, state snapshots, and coverage; the event extension adds a registry and evidence records. Manifests record source and output counts, SHA-256 digests, and acquisition boundaries. Validation covers key uniqueness, referential integrity, coverage states, declared exclusions, and manifest-file agreement.

Coverage states distinguish direct observations, completed queries with no qualifying record, unacquired applicable outcomes, absent mechanisms, and information outside the released scope. This prevents a finite-snapshot nonobservation from being interpreted as a structural zero.

The Solana core contains 832,941 deduplicated terminal outcomes, including 1,651 graduations and 831,290 timeouts, plus 4,953 token-by-horizon transaction proxies and 173,102 decoded swaps for a selected sample of 294 graduated tokens. Base contains 62,618 Clanker launches, the same number of observed Uniswap v4 pool initializations, and 149,921 positive initial-liquidity positions. BNB Chain contains 1,593,679 Four.meme launches and 15,403 observed pools; TRON contains 104,548 SunPump launches and 1,831 observed pools. The latter three are launch universes within their observation windows, not complete terminal-outcome cohorts.

Table 2 reports all windows in UTC while retaining chain-specific acquisition rules. Solana and Base use represented event dates, and TRON is bounded by confirmed pagination exhaustion. BNB membership is defined by verified blocks from manager deployment through the finalized August 12, 2026 snapshot; records with zero or inconsistent timestamps are excluded from temporal analyses.

B Event Timing

Event eligibility requires a named intervention, first-party or onchain activation evidence, a stable outcome, an admissible comparison rule, and a concurrent-shock record. The registry contains four candidates and six evidence records. Base is accepted for a first-observed module-adoption reconstruction. PumpSwap remains conditional because post-migration activity is observed but the exact activation transaction is not verified; BNB Chain and TRON are rejected for the present causal analysis.

A separate Solana creator-fee analysis verifies economic activation on May 13, 2025 at 11:27:06 UTC. A May 12 transaction establishes program support but records zero creator-vault transfer; the next-block transfer after the May 13 upgrade is 10,732 lamports. The analysis does not enter the registry as an accepted comparison design because concurrent product changes and displacement leave no admissible platform control. Clean windows use the verified activation timestamp.

C Stakeholder Metrics

Creator metrics cover participation, graduation, time to graduation, and gross receipts. Trader metrics cover breadth, persistence, concentration, decoded volume, execution quality when available, and source-coded risk. Platform metrics capture revenue and sustained venue activity; community and host-ecosystem metrics capture collected social attention, onchain activity, and spillovers. Each metric retains its unit, horizon, source boundary, coverage state, and interpretation. Mechanical, behavioral, and welfare incidence remain distinct. Figure 6a reports selected Solana measurements.

D Empirical Designs

D.1 PumpSwap

The market panel contains Pump ecosystem, Raydium, Orca, and Meteora daily volume over a ± 90 -day window around March 20, 2025. The outcome is $\log(1 + x_{it})$, where x_{it} is daily protocol volume. L0 compares the Pump ecosystem mean before and after the event. The controlled specification is

$$y_{it} = \alpha_i + \lambda_t + \beta(\text{Pump}_i \times \text{Post}_t) + \varepsilon_{it},$$

where α_i and λ_t are protocol and date fixed effects. The dynamic analogue replaces the post indicator with event-time indicators. Pre-event coefficients diagnose instability. With four protocol clusters, exact Rademacher inference enumerates all $2^4 = 16$ sign assignments. Its two-sided p value is the share of null bootstrap estimates at least as large in absolute value as the observed estimate. Figure 3 displays the 2.5th-to-97.5th percentile span of these 16 null estimates, $[-1.342, 1.342]$ (width 2.685), as a diagnostic reference rather than a population confidence interval.

The 2026 Solana token cohort is not used to estimate the 2025 migration effect. It validates the later venue mechanism. Remote procedure call (RPC) windows count observed pool transactions for graduated tokens at fixed horizons. Moralis records decoded swaps and United States dollar volume for a selected, page-capped subset. We report full-cohort lower bounds separately from complete-window summaries and never treat the decoded subset as representative without qualification.

Figures 5 and 6 summarize the claim-boundary diagnostics reported in the main text.

D.2 Social Metadata

The analysis uses all 832,941 terminal Solana outcomes. The primary outcome is graduation and the exposure is observed Telegram metadata. A linear probability model includes launch-day fixed effects and available token covariates. The matched design forms supported cells from launch day, Twitter presence, website presence, initial-market-capitalization decile, and description length. It retains 20,227 of 20,270 Telegram-present tokens and 586,581 supported controls. The reported average treatment effect on the treated is a matched association because exposure is not assigned.

Uncertainty uses 500 launch-day clustered bootstrap replications. The E-value summarizes the minimum risk-ratio association that an unmeasured confounder would need with both exposure and outcome to explain the observed association under its assumptions. Two falsification exercises constrain interpretation. Outcomes within five minutes test whether the difference predates plausible accumulated Telegram attention, while outcomes after 60 minutes test persistence beyond the immediate launch state. A separate screen evaluates six candidate public shocks for exogenous in-window variation. None passes the event gate.

D.3 Fee Incidence and Base

The creator-fee panel contains 156 platform-day rows, with 21 clean launch days before and after activation for gross outcomes. The anticipation interval begins on May 8, the partial activation day is excluded, and seven-day and 30-day cohort outcomes require complete follow-up. The Pump-only specification is a treated before-and-after contrast. Moonshot remains descriptive because concurrent product changes and displacement violate the causal comparison rule. Weekday-adjusted models use Newey-West standard errors with seven lags; robustness includes a seven-day pretrend and an in-time falsification over eight pseudo-dates.

448 The Base extension identifies the first observed Clanker v4.1 launch with the represented module
449 at block 34,725,785 on August 26, 2025. Six treated launches are matched to six v4.0 controls at
450 three horizons, producing 36 token-horizon rows. This tests event, pool, swap, transfer, and holder
451 reconstruction, but is not powered or sampled for a platform-wide treatment effect.

452 E Monte Carlo Designs

453 Five experiments pair real-data calibration with synthetic assignment or effects. Methods share the
454 generated panel and target estimand; released summaries report Monte Carlo standard errors for
455 rejection, coverage, and bias.

456 **S1, staggered heterogeneity.** Eight Clanker-calibrated adoption cohorts determine timing and
457 target weights; symmetric modifiers generate heterogeneous effects. Static two-way fixed effects and
458 a group-time average treatment effect are compared under zero and positive effects to test whether
459 heterogeneous timing necessarily creates aggregate bias.

460 **S2, announcement and activation.** The observed five-day gap calibrates treatment miscoding
461 for a 0.15-log-point effect. Arms vary coded timing and anticipation. Announcement and verified-
462 activation estimates use the prescribed lag-seven heteroskedasticity-and-autocorrelation-consistent
463 gate; outcomes are bias, coverage, false activation, and true-positive behavior.

464 **S3, four-cluster inference.** Four Solana protocol backbones calibrate residual dependence and
465 relative scale. Ten thousand replications per arm compare cluster-robust variance using normal and
466 $t(3)$ references, exact sign enumeration, and four-assignment randomization inference. Minimum
467 attainable two-sided p values expose discreteness in the decision map.

468 **S4, endogenous adoption.** Clanker creator growth calibrates latent adoption selection. Zero and
469 positive effects are crossed with independent, moderate, and strong selection. Static two-way fixed
470 effects, an event-time-zero group-time estimator, and sensitivity intervals are compared on bias,
471 coverage, width, and decisions under the registered deviation bound $\bar{M}$.

472 **S5, temporal aggregation.** The March 20 PumpSwap date fixes Thursday event geometry, with
473 transient and persistent effects placed across alternative weekdays. Daily, event-aligned weekly, naive
474 calendar-week, and exposure-weighted estimators target the same seven-day effect where possible,
475 separating point-estimate disagreement from a decision flip.

476 F LLM Boundary-Adherence Evaluation

477 The frozen PumpSwap V2 audit contains 714 successful calls, 238 per backend: 30 cumulative
478 L0–L2 calls, 160 complete M4–M7 factorial calls, 24 answer-leakage controls, and 24 evidence-free
479 controls. Returned identifiers are gpt-5.6-terra, deepseek-v4-pro, and qwen3:14b; the local
480 Qwen digest begins bdbd181c33f2. Temperatures are zero. Five DeepSeek outputs required format-
481 only repair; none failed, and GPT and Qwen required no repair. For a given factor, the other three
482 factorial bits define $2^3 = 8$ matched backgrounds per backend. These backgrounds are the primary
483 design variation, while the 10 repeated calls per cell measure response stability at zero temperature.
484 The primary factor intervals use 2,000 seeded hierarchical draws. Within each backend, a draw
485 resamples the eight backgrounds and then resamples calls within the matched factor-present and
486 factor-absent cells. Backends form a fixed-design, equally weighted panel and are not resampled.
487 The original call-level bootstrap is retained as a descriptive sensitivity analysis; neither interval is
488 population-model inference.

Table 5: PumpSwap evidence modules in the frozen V2 audit. The descriptions follow the prompt slot catalogue, the evidence-block builder, and the archived evidence blocks. M0–M3 form the fixed analytical background in the factorial arm, while M4–M7 vary independently. These prompt components draw on the evidence ladder but are not identical to its analytical rungs.

Module	Evidence disclosed	Role in the audit	Factor status
M0	Treated-series before and after estimate with the case date	Establishes the initial operational observation without a comparison series	Fixed
M1	Naive control-group difference-in-differences estimate	Adds the observed difference between treated and comparison series	Fixed
M2	Unit and date fixed-effects estimate	Adds the adjusted market-level contrast while retaining an aggregate-volume outcome	Fixed
M3	Average dynamic postevent estimate from the single-event analogue	Adds dynamic evidence before the preperiod diagnostic is disclosed	Fixed
M4	Event-study coefficients and a neutral preperiod summary	Tests whether instability before the event constrains a causal interpretation	Varied
M5	Stakeholder-disaggregated metric battery with units and value labels	Tests whether the response separates stakeholder evidence from aggregate market activity	Varied
M6	Exact Rademacher inference over four clusters and 16 sign assignments	Tests whether design-specific uncertainty changes market-causal affirmation	Varied
M7	Frequency sensitivity and aggregate-artifact availability	Discloses aggregation dependence and unresolved coverage or scope	Varied

Under complete M0–M7 disclosure, GPT and DeepSeek return the correct boundary in 10/10 calls, whereas Qwen does so in 0/10 and makes an unsafe causal affirmation in 10/10. M6 is the only factorial module with a nonpositive unsafe-affirmation contrast for every backend.

The targeted Telegram second-case evaluation contributes 60 additional successful calls under the same three-backend panel and zero-temperature setting, for 774 calls across the two current audits. T0 supplies the observed and matched association, clustered interval, E-value, and matched support. T1 adds the five-minute falsification, the 60-minute result, the 0/6 qualifying-shock screen, and the unavailable or proxied controls. The case-specific output schema records causal status separately from predictive-association status. It does not include the archived reference-boundary text in either prompt.

Table 6: Telegram targeted second-case evaluation. Entries are successful responses satisfying each binary metric out of 10 registered calls. T1-minus-T0 is zero for every model and metric.

Backend	Condition	Unsafe causal	Correct boundary	Predictive retained
GPT-5.6 Terra	T0 association	0/10	10/10	10/10
GPT-5.6 Terra	T1 boundary-complete	0/10	10/10	10/10
DeepSeek-V4-Pro	T0 association	0/10	10/10	10/10
DeepSeek-V4-Pro	T1 boundary-complete	0/10	10/10	10/10
Qwen3-14B	T0 association	0/10	10/10	10/10
Qwen3-14B	T1 boundary-complete	0/10	10/10	10/10

All T0 responses cite T0 and mark T1 as missing, whereas all T1 responses cite both blocks without missing slots. The ceiling precludes estimating disclosure improvement, although T1 claims use the additional evidence. The 774-call archive provides manifests, evidence blocks, registries, parsed outputs, scores, and summaries. Its PumpSwap release separates the post-run code snapshot from the historical commit and records the absence of raw provider JSON. The 80-response deepseek-v4-flash pilot remains provenance bounded. These two cases examine causal escalation from observational and predictive evidence within one scaffold family. They do not establish broad generalization, autonomous causal identification, or population model rankings.